Note

Recall that subtraction on numbers is not commutative. Is set difference commutative? We expect that the set difference is not commutative as well. For instance, consider $A=\{a,b,c\}$, $B=\{b,c,d\}$. $A-B=\{a\}$, $B-A=\{d\}$; we see that $A-B\neq B-A$.

1.6.2 Associative Property

Now, we perform operations on union and intersection for three sets.

Let $A=\{-1,0,1,2\}$, $B=\{-3,0,2,3\}$ and $C=\{0,1,3,4\}$ be three sets.

Now, $B\cup C = \{-3,0,1,2,3,4\}$

$A\cup(B\cup C) = \{-1,0,1,2\}\cup\{-3,0,1,2,3,4\}$

$= \{-3,-1,0,1,2,3,4\}$ ... (1)

Then, $A\cup B = \{-3,-1,0,1,2,3\}$

$(A\cup B)\cup C = \{-3,-1,0,1,2,3\}\cup\{0,1,3,4\}$

$= \{-3,-1,0,1,2,3,4\}$ ... (2)

From (1) and (2), $A\cup(B\cup C)=(A\cup B)\cup C$.

This is associative property of union among sets A, B, and C.

Now, $B\cap C = \{0,3\}$

$A\cap(B\cap C) = \{-1,0,1,2\}\cap\{0,3\}$

$= \{0\}$ ... (3)

Then, $A\cap B = \{0,2\}$

$(A\cap B)\cap C = \{0,2\}\cap\{0,1,3,4\}$

$= \{0\}$ ... (4)

From (3) and (4), $A\cap(B\cap C)=(A\cap B)\cap C$.

This is associative property of intersection among sets A, B and C.

Associative property: For any three sets A, B and C

(i) $A\cup(B\cup C)=(A\cup B)\cup C$ (ii) $A\cap(B\cap C)=(A\cap B)\cap C$

Example 1.20 If $A=\left\{-\frac{1}{2},0,\frac{1}{4},\frac{3}{4},2\right\}$, $B=\left\{0,\frac{1}{4},\frac{3}{4},2,\frac{5}{2}\right\}$ and $C=\left\{-\frac{1}{2},\frac{1}{4},1,2,\frac{5}{2}\right\}$, then verify $A\cap(B\cap C)=(A\cap B)\cap C$.

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